MM7025 HW1

Lex: lexicographic order

GLex: First by higher total degree, then by Lex

GRevLex: —"—, but reverse Lex

2. a) $f(x,y) = x^7y^2 + x^3y^2 - y + 1$
$g(x,y) = xy^2 - x$
} same for all orderings

$$
\begin{array}{r|l}
 & x^6 + x^2 \\
\hline
xy^2 - x & x^7y^2 + x^3y^2 - y + 1 \\
 & -x^7y^2 + x^7 \\
\hline
 & x^3y^2 + x^7 - y + 1 \\
 & -x^3y^2 + x^3 \\
\hline
 & x^7 + x^3 - y + 1
\end{array}
$$

Remainder $x^7 + x^3 - y + 1$, for all orderings

b) f same as above.

$g(x,y) = x - y^3$, in Lex

$g(x,y) = -y^3 + x$, in GLex, GRevLex

Lex first:

$$
\begin{array}{r|l}
 & x^5y^2 + x^4y^5 + x^3y^8 + x^2y^{11} + x^2y^2 + xy^{14} + xy^5 \\
 & \quad + y^{17} + y^8 \\
\hline
x - y^3 & x^7y^2 + x^3y^2 - y + 1 \\
 & -x^7y^2 + x^5y^5 \\
\hline
 & x^5y^5 + x^3y^2 - y + 1 \\
 & -x^5y^5 + x^4y^8 \\
\hline
 & x^4y^8 + x^3y^2 - y + 1 \\
 & -x^4y^8 + x^3y^{11} \\
\hline
 & x^3y^{11} + x^3y^2 - y + 1 \\
 & -x^3y^{11} + x^2y^{14} \\
\hline
 & x^3y^2 + x^2y^{14} - y + 1 \\
 & -x^3y^2 + x^2y^5 \\
\hline
 & x^2y^{14} + x^2y^5 - y + 1 \\
 & -x^2y^{14} + xy^{17} \\
\hline
 & x^2y^5 + xy^{17} - y + 1 \\
 & -x^2y^5 + xy^8 \\
\hline
 & xy^{17} + xy^8 - y + 1 \\
 & -xy^{17} + y^{20} \\
\hline
 & xy^8 + y^{20} - y + 1 \\
 & -xy^8 + y^{11} \\
\hline
 & y^{20} + y^{11} - y + 1
\end{array}
$$

Should be $y^{23} + y^{11} - y + 1$, but close enough.

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2.b) $g(x,y) = -y^3 + x$, in GLex, GRevLex

$-y^3 + x \overline{)\, x^7y^2 + x^3y^2 - y + 1}$

x^7y^2 indivisible by $-y^3 \Rightarrow$ remainder $f(x)$

c) Lex $\Rightarrow$ $r_1 = y^{23} + y^{11} - y + 1$, from b)

$x - y^3$ cannot divide this, so remainder is r_1

GLex, GRevLex: $r_1 = f$, from b) above

Then $r_2 = x^7 + x^3 - y + 1$, from a)

Now r_2 is not divisible by $-y^3 + x$,

so remainder is r_2.